

ATHARVA HEMANT JOSHI

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EDUCATION

B.E., Computer Engineering - Pune Institute of Computer Technology (PICT)

Aug 2024 – Jul 2027

Diploma, Information Technology - Government Polytechnic, Nagpur

Sep 2021 – May 2024

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, SQL, PHP

Frontend: React.js, Angular, Next.js, HTML5, CSS3, REST API Integration, WebSockets, Browser Fundamentals

Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices

Java/J2EE: Core Java, OOP, JDBC, Servlets, RESTful Web Services, Android SDK

AI/ML: LLMs, RAG, YOLO, OpenCV, Scikit-learn, TensorFlow, MLflow

Databases: MySQL, PostgreSQL, MongoDB, Redis, SQL

Cloud/DevOps: AWS, Docker, CI/CD, Git, Linux

CS Fundamentals: DSA, OOP, DBMS, OS, Computer Networks, System Design, Design Patterns

EXPERIENCE

Computer Vision & ML Intern | *Thelios AI*

Jun 2025 – Sep 2025

Cut image-preprocessing latency 35% as measured by pipeline throughput benchmarks by engineering batched, parallelized Python ETL pipelines with REST-controlled orchestration for large-scale image datasets.

Improved YOLO detection mAP 18% over baseline as measured on held-out validation sets by fine-tuning models with custom augmentation, MLflow experiment tracking, and unit-tested reproducible builds.

Web Developer Intern | *MuscleBox*

May 2024 – Aug 2024

Designed and developed the official website for Muscle Box, a corporate fitness brand, with a focus on **responsive UI, user-friendly features, and brand consistency**. Implemented web solutions aligned with the company's **promotional and business requirements**.

PROJECTS

ISRO IRoC-U Robotics Challenge 2025 - Autonomous Drone Mission Control

AIR 1 / 500+ Teams

Python · Flask · OpenCV · YOLO

Trained lightweight YOLOv8 models on edge hardware (Raspberry Pi) for real-time detection of landing pads and boundary markers at ≥ 15 FPS.

Delivered sub-second telemetry latency as measured by live dashboard responsiveness by architecting a Flask REST API backend with WebSocket streaming and modular OpenCV landing-zone detection.

Adversia - AI Red Team-Blue Team Payment Security Lab

Mastercard Hackathon

Python · NumPy/Pandas · Adversarial ML · TreeSHAP · Next.js · Docker

Achieved 93.5% ROC-AUC and 61.1% recall at 1% fraud prevalence as measured on 361K simulated transactions (2,061 customers, 1,327 merchants) by implementing Histogram-GBDT and Random-Forest detectors across 7 fraud families and 83 causal behavioral features.

Improved held-out attack recall 13.5pp (22.0%→35.5%) and F1 0.357→0.519 at matched 0.991% FPR as measured by adversarial evaluation by designing an adaptive Red-Team↔Blue-Team retraining loop, verified with 92/92 unit and 14/14 E2E tests.

Delivered explainable fraud decisions at 2.7×10^{-8} TreeSHAP attribution error as measured against exact Shapley values by integrating model-explainability tooling into the live detection pipeline.

SovereignGuard - Secure & Sovereign LLM Data Pipeline

B.E. Capstone, Ongoing

Python · Docker · LLMs · RAG · LoRA/PEFT · Data Privacy & Governance

Enforced full tenant isolation and data-privacy compliance as measured by ingestion-audit coverage by designing classification, PII-detection, and anonymization workflows for proprietary enterprise data.

Eliminated dependency on external model APIs as measured by fully self-hosted inference and fine-tuning by building organization-specific RAG and LoRA/PEFT workflows on open-weight LLMs.

Green Model Advisor - Carbon-Aware AI Inference Router

Python · FastAPI · Docker · CodeCarbon

Cut compute usage 45% versus static model selection as measured by CO₂-tracking benchmarks by building a FastAPI microservice router that dynamically selects LLMs on latency, accuracy, cost, and emissions.

ISRO Bharatiya Antariksh AI/ML Hackathon 2025 - Satellite Air-Pollution Analytics

Top 28 / 8,744 Teams

Hit 88% prediction accuracy with F1 improved 15% as measured on satellite AQI held-out data by engineering a 5-year multi-source ingestion pipeline and distilling 12 high-signal features from 40+ raw variables.

Open Source: gVisor | eksctl (Amazon EKS) | SoL-Pi — Resolved GitHub issues, debugged existing codebases, implemented fixes, added/validated tests, and contributed through structured pull requests and code-review workflows.

ACHIEVEMENTS & LEADERSHIP

AIR 1, ISRO IRoC-U Robotics Challenge 2025 - 1st nationwide of 500+ teams (incl. IITs, NITs) for GNSS-free autonomous drone.

AIR 9, DD Robocon India 2026 - 100/100 in Stage 1 & Stage 2; Best Technical Report Award.

Finance & Project Management Lead, PICT Robotics (2026) - owns architecture, task pipelines, and integration milestones for a 30-member team.